

Edge AI : Real-Time Facial Recognition and Behavior Aware Surveillance Using Raspberry Pi and XAI

An Edge-AI surveillance system using Raspberry Pi and a camera to detect and recognize people, track movements, identify abnormal activities, provide explainable AI-based alerts, and store event details locally in real time

Project & Hardware Requirements

1. **Project:** Edge-AI based real-time facial recognition and behavior-aware surveillance using Raspberry Pi and XAI.
2. The system will process **live camera footage directly on the Raspberry Pi** without depending on cloud processing.
3. It will detect people, recognize faces, track movement, and identify abnormal activities.
4. Abnormal activities include **unauthorized entry, restricted-zone access, loitering, and unusual movements**.
5. **Raspberry Pi 5 – 8GB** is required as the main AI processing unit.
6. **Raspberry Pi Camera Module 3** is required for real-time video input.
7. **64/128GB MicroSD card** is required for OS, AI models, software, logs, and event data.
8. **27W USB-C power supply** is required for stable Raspberry Pi operation.
9. **Raspberry Pi 5 Active Cooler** is required for continuous AI processing.
10. **Raspberry Pi 5 Case** is required for safe physical deployment.
11. **Micro-HDMI to HDMI cable** is required for display/setup and debugging.
12. **Ethernet cable** is required for network connectivity, configuration, and dashboard/API access.
13. The system will generate **real-time alerts and store event images, timestamps, and details**.
14. **XAI** will explain the reason behind detected abnormal events, making the system transparent and understandable.